

Robust Adjoint-Optimized 2×2 MMI 3-dB Coupler

openEBL 2026-10 design document

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1. Objective

A compact ($3.0 \times 10.5 \mu\text{m}^2$) paired-interference 2×2 MMI 3-dB coupler on 220-nm SOI. The MMI sidewall (12-point PCHIP spline, mirror-symmetric) was shape-optimised with a multi-fidelity adjoint method: 2.5D varFDTD gradients corrected by 3D FDTD output space mapping, with five fabrication corners (etch bias ± 20 nm, Si 210/230 nm) inside one robust figure of merit. The chip measures the splitting precision of the robust coupler and of a conventional rectangular MMI of the same footprint through unbalanced Mach-Zehnder interferometers (MZIs).

2. Simulated performance (3D FDTD, 20 nm mesh, C-band 1530–1565 nm)

Metric	Conventional MMI	Robust adjoint MMI
Worst IL, nominal (dB)	0.502	0.176
Worst imbalance , nominal (dB)	0.340	0.115
0.3-dB bandwidth, nominal (nm)	0	81
Worst imbalance over 7 fab corners (dB)	0.413	0.191
Worst IL over 7 fab corners (dB)	0.583	0.315
Worst-case balanced-MZI extinction ratio (dB)	26.5	33.1
Phase error, all corners	$\leq 2.9^\circ$	$\leq 2.1^\circ$

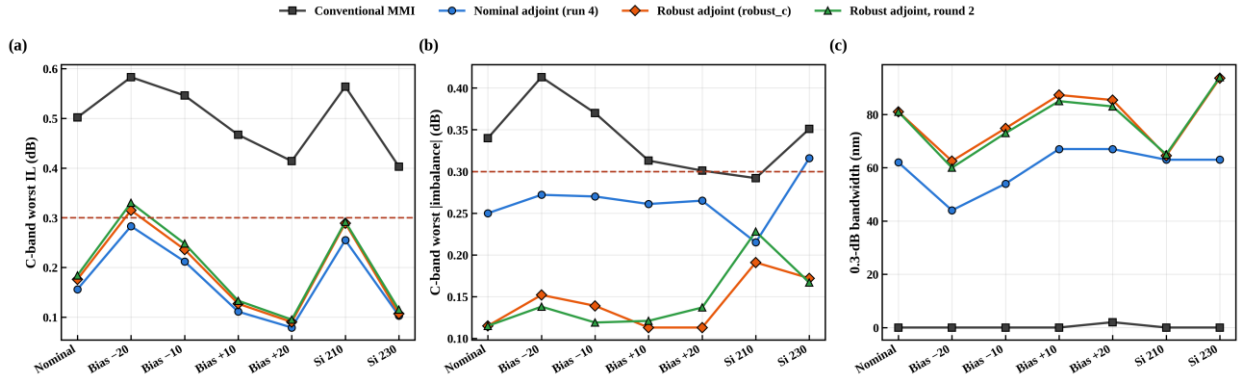


Figure 1. Simulated C-band worst-case insertion loss, imbalance and 0.3-dB bandwidth versus etch bias and Si thickness.

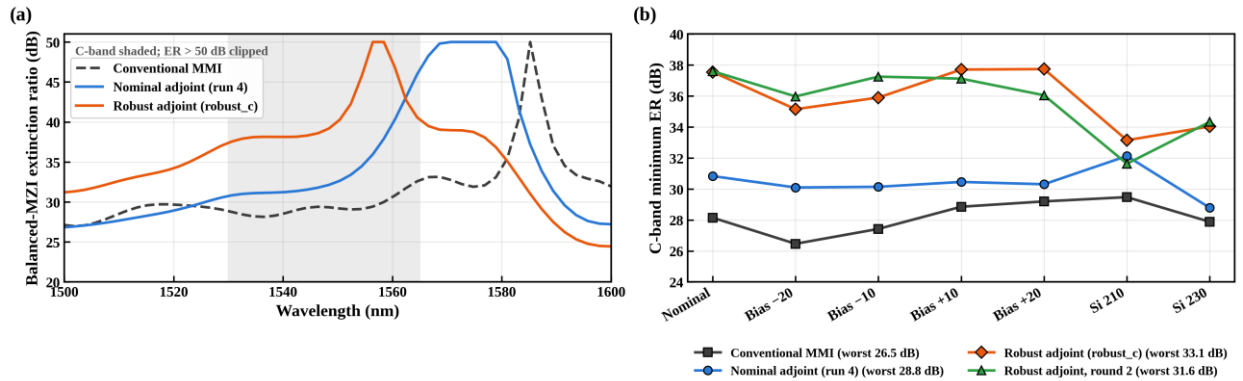


Figure 2. Simulated extinction ratio of a balanced MZI built from two couplers, nominal (a) and across fabrication corners (b).

3. Layout

- TE 1550 grating couplers (GC_TE_1550_8degOxide_BB), 127 μm pitch, laser on the middle GC, detectors above and below (DFT rules).
- Unused MMI input terminated (ebeam_terminator_te1550). Waveguides: Strip TE 1550 nm, $w = 500$ nm, bend radius 5 μm .

- Smallest feature: 150-nm gap between the input tapers at the MMI facet. SEM request boxes (layer 200/0, 4:3) can be placed on the MMI bodies.

Circuit (opt_in label)	GCs	Purpose
MZIrobust_c	3	Two robust MMIs, $\Delta L \approx 91 \mu\text{m}$ (FSR $\approx 6 \text{ nm}$); fringe extinction gives the split ratio independent of GC variations
MZI_conventional	3	Same MZI with the conventional rectangular MMI (reference)

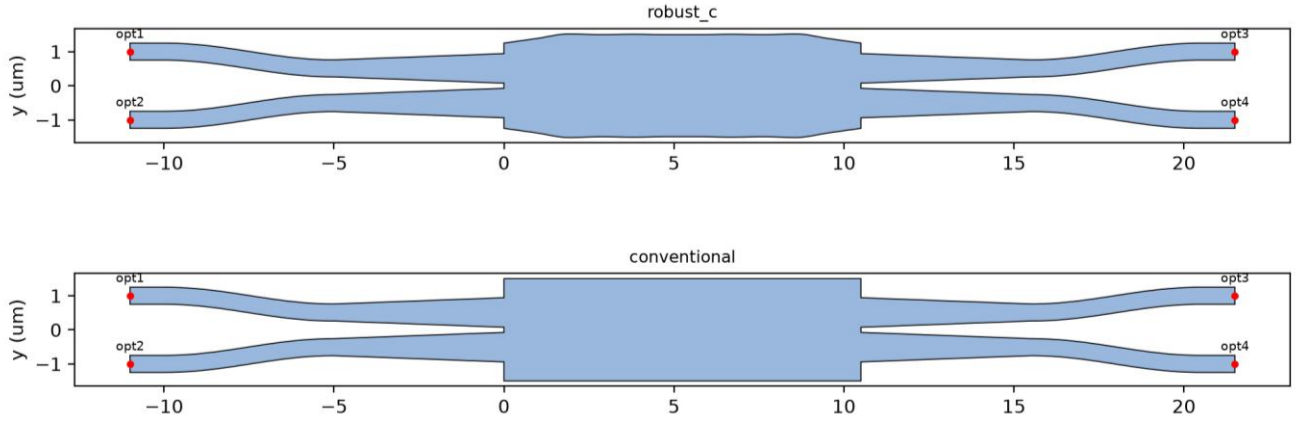


Figure 3. Robust and conventional MZIs

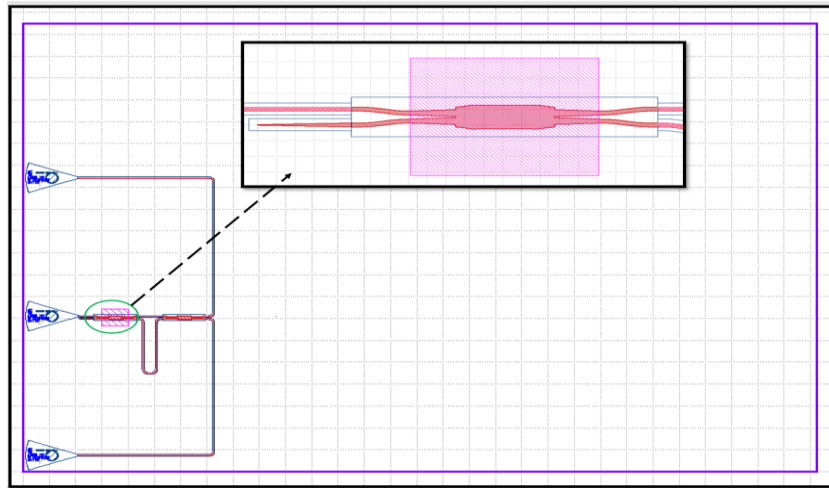


Figure 4. KLayout view of the robust MZI: input GC, terminated port 2, two robust MMIs, straight short arm and U-shaped long arm.

4. Test plan and analysis

- Laser 1500–1600 nm, TE, both MZI outputs recorded.
- For two identical couplers with power coupling r , the MZI extinction ratio is $\text{ER} \approx -20 \log_{10}|1 - 2r|$, so the fringe depth gives r without grating-coupler calibration.
- Expected: robust MZI $\text{ER} \geq 33 \text{ dB}$ versus $\approx 26\text{--}29 \text{ dB}$ for the conventional MZI across the C-band.
- The FSR gives the group index; the wavelength dependence of ER gives the coupler bandwidth.